

GRID-CLOCK COSMOLOGY

PAPER III OF III | STRONG GRAVITY

Singularity Avoidance by Saturated Grid Response

*Regular Black-Hole Cores, Horizon Structure, and Conditional Strong-Gravity
Signatures*

NS

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Revised and consolidated three-paper edition

Abstract

We present the strong-gravity paper of the consolidated Grid-Clock Cosmology (GCC) series. The successful weak-field and cosmological correspondence limits require ordinary general relativity in nonextreme environments, and Paper II constrains the homogeneous M -- V exchange to the immediate neighbourhood of its exact $n = 3$ Λ CDM limit. We therefore keep $n = 3$ and isolate a local regime in which the coarse-grained grid response saturates at extreme curvature.

Because the microscopic saturation law is not yet known, the calculable completion adopts the effective mass profile

$$M_{\text{eff}}(r) = M \frac{r^3}{r^3 + r_c^3},$$

which gives a Hayward-type static spherical geometry. The exterior approaches Schwarzschild and the centre is de Sitter-like. For every nonzero core radius r_c , the effective density, Ricci scalar, Ricci-tensor square, and Kretschmann scalar remain finite. Defining $\epsilon = r_c/r_s$, we derive the two-horizon, extremal, and horizonless branches with

$$\epsilon_{\text{crit}} = \left(\frac{4}{27}\right)^{1/3},$$

and obtain correlated shifts in the photon sphere, shadow radius, innermost stable circular orbit, Hawking temperature, and eikonal ringdown proxies. A regular centre does not by itself predict gravitational-wave echoes; echoes require an additional partially reflecting near-horizon or horizonless layer.

The paper proves local curvature regularity for the chosen saturation ansatz but does not prove global geodesic completeness, inner-horizon stability, a rotating solution, or unitary evaporation. Variable effective grid number, directional cosmology, and cyclic evolution are mentioned only as future cross-sector extensions and are not used in the black-hole calculations. The result is a falsifiable singularity-avoidance candidate, not a completed quantum theory of black holes.

Keywords: emergent spacetime; discrete quantum gravity; singularity avoidance; regular black holes; saturation core; black-hole shadow; ISCO; ringdown; gravitational-wave echoes; black-hole thermodynamics.

Status Note

This manuscript is a theoretical effective-model paper. The regular metric is a Hayward-type saturation ansatz motivated by bounded grid response. It is not a unique nonlinear solution derived from the microscopic GCC update rule, and the static spherical model is not a substitute for a rotating astrophysical black-hole calculation.

1 Introduction

General relativity describes weak and strong gravitational phenomena with extraordinary success, yet its classical black-hole solutions contain curvature singularities where the continuum description becomes incomplete. Singularity theorems clarify the generic character of geodesic incompleteness under broad assumptions, but they do not specify the microscopic replacement of the singular region. Regular black-hole geometries provide one controlled way to study this question: the exterior can remain close to Schwarzschild or Kerr while the center is replaced by a finite-density core.

Grid-Clock Cosmology treats physical distance, proper time, and gravitational geometry as emergent coarse-grained properties of a discrete substrate. The underlying graph has a fixed set of adjacency relations and evolves through a universal integer-valued update ordering. In ordinary regimes, the coarse-grained lapse, shift, and spatial metric must reproduce local Lorentz invariance and general relativity. Paper I deliberately left the nonlinear strong-field response open. Paper II showed that the simplest homogeneous cosmological exchange is not detected: present background, CMB, growth, RSD, and lensing data drive the isotropic sector close to its exact Λ CDM limit. The natural remaining question is therefore not whether a large homogeneous exchange replaces standard cosmology, but whether bounded response becomes relevant only in extreme local curvature.

The present paper addresses that question with a deliberately minimal construction. We introduce a bounded grid-response order parameter, state a candidate effective action, and adopt a one-parameter regular mass profile. The metric profile itself is not new: it is of Hayward type and belongs to the established regular-

black-hole literature. The GCC contribution is the saturation interpretation, its embedding in the previous correspondence hierarchy, the no-retuning observable programme, and an explicit separation between derived results, ansatz-dependent predictions, and unresolved microscopic dynamics.

The concrete objectives are:

1. Define a strong-field saturation variable without modifying the successful weak-field and cosmological correspondence limits.
2. Construct a regular static spherical metric and prove that its central density and polynomial curvature invariants are finite.
3. Derive one-parameter predictions for shadows, circular orbits, ISCO frequencies, thermodynamics, and an eikonal ringdown diagnostic.
4. State precisely what singularity removal does and does not solve about black-hole information.
5. Provide falsification conditions and a development path toward rotation, collapse, evaporation, and full perturbative stability.

2 Relation to the GCC Series and Theoretical Status

2.1 Correspondence hierarchy

The strong-field model is constrained by the correspondence structure established in the preceding GCC papers. The microscopic grid and universal update index are foundational assumptions. Local Lorentz invariance, unitary low-energy quantum evolution, and weak-field general relativity are mandatory correspondence requirements. The homogeneous cosmological sector has an exact Λ CDM point and is observationally driven to its immediate neighborhood. Black-hole regularization belongs to a higher, completion-dependent level: it cannot be inferred from the weak-field expansion alone.

Table 1. Correspondence hierarchy for the GCC strong-gravity sector.

Level	Content	Status in this paper
I	Discrete graph and ordered microscopic updates	Inherited foundational postulate
II	Local Lorentz symmetry, quantum and GR correspondence	Mandatory low-energy limit
III	Homogeneous M--V cosmology and perturbations	Constrained near the $n=3$ CDM limit
IV	Bounded nonlinear response and regular black-hole core	Effective completion candidate tested here

2.2 No-retuning principle

Once the static mass profile is selected, the single ratio $\epsilon = r_c/r_s$ must determine the horizon positions, photon sphere, shadow, ISCO, surface gravity, and eikonal ringdown indicators simultaneously. It is not permissible to choose a different core radius for each observable. A future rotating extension may introduce the spin parameter a_* , but the same strong-field parameters must then be shared across electromagnetic and gravitational-wave tests.

2.3 What is and is not derived

Table 2. Status of derived, ansatz-dependent, and open claims.

Claim	Status
Finite central curvature for the adopted mass profile	Derived exactly
Two-horizon, extremal, and horizonless branches	Derived exactly
Shadow, ISCO, and temperature trends	Derived within the static ansatz
Bounded grid response as the physical origin of the core	Interpretive completion hypothesis
Covariant scalar action generating the mass profile	Candidate action; not yet uniquely solved
Rotating astrophysical black-hole metric	Open
Unitary Hawking evaporation and Page curve	Open
Dynamical stability of the inner horizon	Open and potentially decisive

3 Bounded Grid Response and a Candidate Effective Action

3.1 Saturation variable

Let $C(x)$ denote a coarse-grained local capacity of the grid to support stable causal, geometric, and field response, and let C_0 be its reference-vacuum value. We define the dimensionless response-deficit variable

$$\chi(x) = 1 - \frac{C(x)}{C_0}, \quad 0 \leq \chi < 1 \quad (3.1)$$

The weak-field regime has χ much smaller than unity. Strong curvature may drive χ toward a saturation layer. The bound is not interpreted as a material medium or a preferred-frame ether. It is a covariant order parameter describing the breakdown of the linear constitutive relation between grid state and emergent geometry.

3.2 Candidate covariant action

One effective realization is a scalar response field coupled weakly to ordinary matter and more strongly to curvature invariants in the nonlinear regime:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{Pl}^2}{2} R - \frac{\Lambda_\chi^2}{2} K(\chi) (\nabla\chi)^2 - U(\chi) + \alpha_\chi F(\chi) \mathcal{G} \right] + S_m[\psi, A^2(\chi) g_{\mu\nu}] \quad (3.2)$$

$$\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \quad (3.3)$$

Here $K(\chi) > 0$ is required to avoid a wrong-sign kinetic mode, $U(\chi)$ enforces a bounded response region, $F(\chi)$ controls strong-curvature excitation through the Gauss-Bonnet invariant, and $A(\chi)$ must approach unity rapidly in ordinary environments to suppress fifth forces and equivalence-principle violations.

$$U(\chi) = \frac{1}{2} m_\chi^2 \Lambda_\chi^2 \chi^2 + \Lambda_{\text{sat}}^4 \frac{\chi^4}{(1-\chi)^2} \quad (3.4)$$

Equation (3.2) is not claimed to be unique. More importantly, the metric studied below has not yet been derived as a unique exact solution of this action. The action states the type of nonlinear completion required; the mass profile provides a calculable strong-field ansatz.

3.3 Weak-field recovery

A viable completion must satisfy $K \rightarrow 1$, $A \rightarrow 1$, $F \rightarrow 0$, and $U \rightarrow 0$ as $\chi \rightarrow 0$. In this limit the Einstein-Hilbert term dominates, ordinary matter follows the same metric, and the external solution of an isolated compact object approaches the Schwarzschild geometry. The cosmological background remains the near- $n=3$ solution already constrained in Paper II; the saturation parameter introduced here is not used to refit the homogeneous universe.

4 Minimum Static Saturation-Core Geometry

4.1 Metric and effective mass profile

In geometric units $G=c=1$, consider the static spherical line element

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (4.1)$$

$$f(r) = 1 - \frac{2M_{\text{eff}}(r)}{r}, \quad M_{\text{eff}}(r) = \frac{Mr^3}{r^3 + r_c^3} \quad (4.2)$$

The ADM mass is M and r_c is the saturation-core radius. Restoring constants, the Schwarzschild radius is $r_s = 2GM/c^2$. Defining $x = r/r_s$ and $\epsilon = r_c/r_s$ gives

$$f(x) = 1 - \frac{x^2}{x^3 + \epsilon^3} \quad (4.3)$$

This form is mathematically Hayward-like. Its use here should not be read as a claim that GCC uniquely predicts the Hayward metric. It is the minimum profile with $M_{\text{eff}} \propto r^3$ at the origin and $M_{\text{eff}} \rightarrow M$ at large distance.

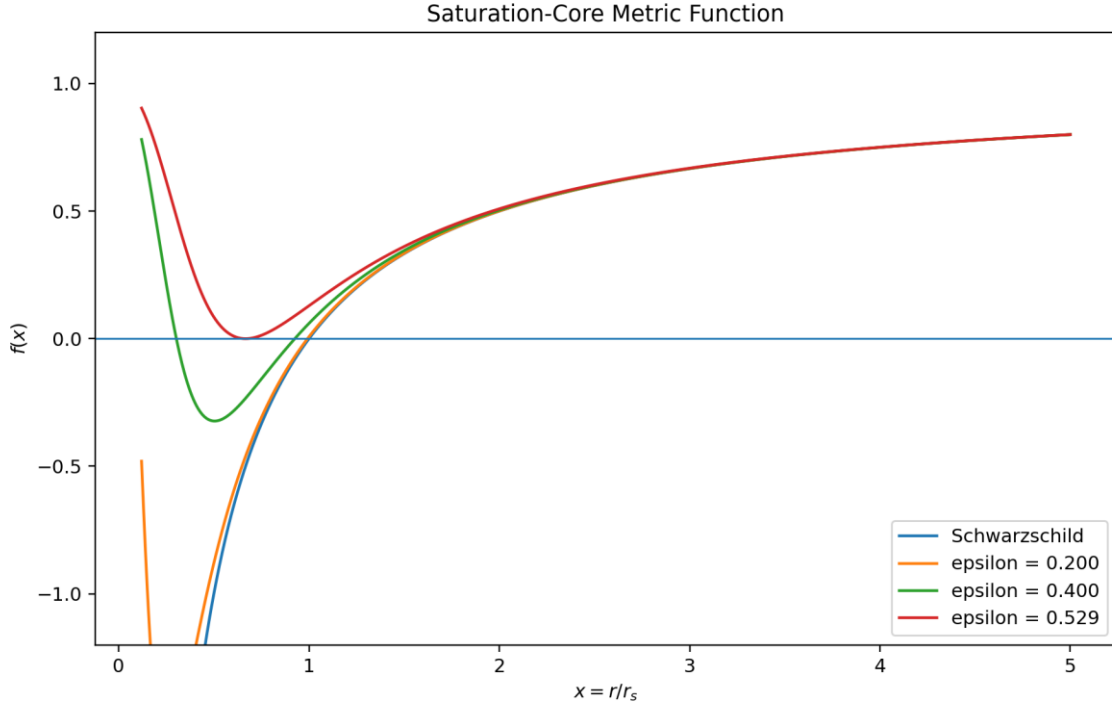


Figure 1. Metric function for the Schwarzschild limit and representative saturation-core ratios. The two horizons merge at the extremal value ϵ_{crit} .

4.2 Exterior and central limits

$$f(x) = 1 - \frac{1}{x} + \frac{\epsilon^3}{x^4} + \mathcal{O}\left(\frac{\epsilon^6}{x^7}\right), \quad x \gg \epsilon \quad (4.4)$$

The leading exterior term is Schwarzschild. The first static correction scales as r^{-4} in f and is therefore concentrated near the horizon.

$$f(r) = 1 - \frac{r^2}{L_c^2} + \mathcal{O}(r^5), \quad L_c^2 = \frac{r_c^3}{r_s}, \quad r \ll r_c \quad (4.5)$$

The center is de Sitter-like with effective cosmological constant $\Lambda_c = 3/L_c^2$. The singular $1/r$ behavior is replaced by a finite-curvature core.

5 Regularity and Effective Stress-Energy

5.1 Finite curvature

For any smooth spherical mass function with $M_{\text{eff}}(r)=O(r^3)$, the metric approaches de Sitter space at the origin and polynomial curvature invariants remain finite. For the profile in Eq. (4.2), the central limits are

$$R(0) = \frac{12r_s}{r_c^3}, \quad R_{\mu\nu}R^{\mu\nu}(0) = \frac{36r_s^2}{r_c^6}, \quad \mathcal{K}(0) = \frac{24r_s^2}{r_c^6} \quad (5.1)$$

Here $\mathcal{K}$ denotes the Kretschmann scalar. All three invariants are finite for nonzero r_c and diverge only in the Schwarzschild limit $r_c \rightarrow 0$.

5.2 Effective anisotropic fluid

Interpreting the geometry through Einstein equations gives an effective anisotropic stress tensor. In geometric units,

$$\rho(r) = \frac{3Mr_c^3}{4\pi(r^3+r_c^3)^2}, \quad p_r(r) = -\rho(r) \quad (5.2)$$

$$p_t(r) = \frac{3Mr_c^3(2r^3-r_c^3)}{4\pi(r^3+r_c^3)^3} \quad (5.3)$$

At the center $p_r=p_t=-\rho$, reproducing a vacuum-energy equation of state. The weak energy condition is satisfied because ρ is nonnegative, $\rho+p_r=0$, and

$$\rho + p_t = \frac{9Mr_c^3 r^3}{4\pi(r^3+r_c^3)^3} \geq 0 \quad (5.4)$$

The strong energy condition is violated in the central region, as expected for a de Sitter core. The dominant energy condition is not globally satisfied by this effective stress tensor; this is an important limitation of the minimal ansatz and a reason to derive the profile from a complete field configuration rather than treat it as fundamental matter.

6 Horizons and Causal Branches

6.1 Horizon equation and critical core

$$f(x_h) = 0 \quad \Leftrightarrow \quad x_h^3 - x_h^2 + \epsilon^3 = 0 \quad (6.1)$$

A double horizon occurs when the polynomial and its derivative vanish. The nonzero double root is $x_{\text{ext}}=2/3$, giving

$$\epsilon_{\text{crit}}^3 = \frac{4}{27}, \quad \epsilon_{\text{crit}} = \left(\frac{4}{27}\right)^{1/3} = 0.529133684 \quad (6.2)$$

Table 3. Horizon branches of the saturation-core geometry.

Core ratio	Causal structure
$0 \leq \epsilon < \epsilon_{\text{crit}}$	Two positive horizons: outer event horizon and inner Cauchy horizon
$\epsilon = \epsilon_{\text{crit}}$	Degenerate extremal horizon at $x=2/3$
$\epsilon > \epsilon_{\text{crit}}$	No horizon; regular compact object rather than a black hole

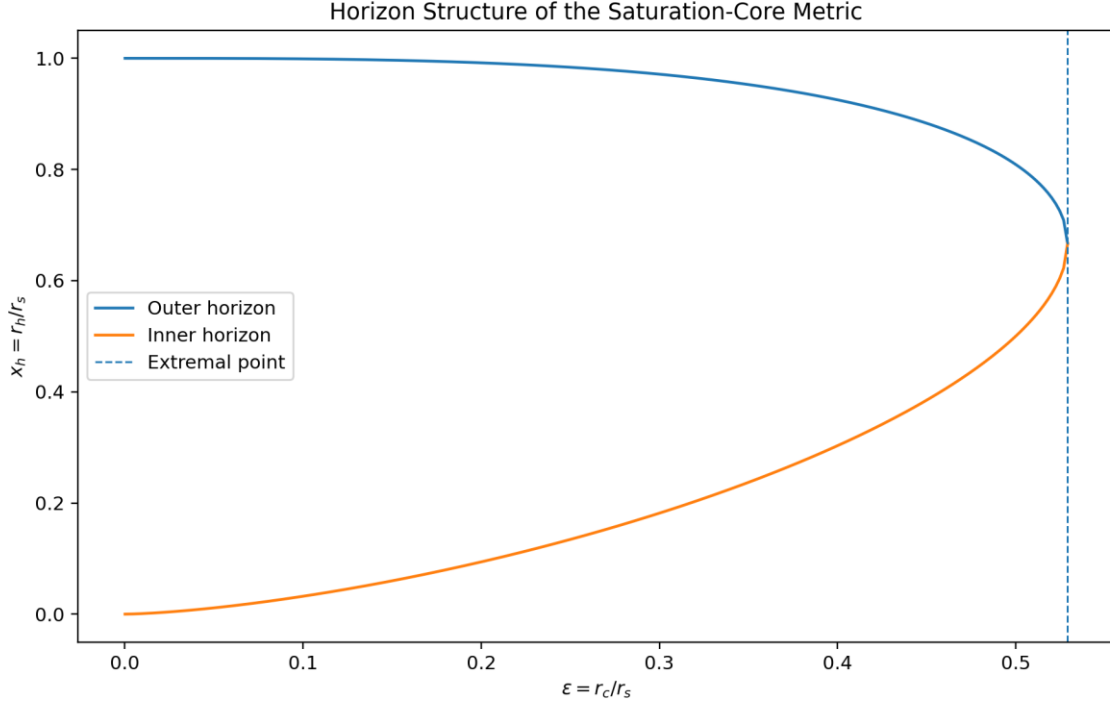


Figure 2. Inner and outer horizon branches. The branches merge at $\epsilon_{\text{crit}} = (4/27)^{1/3}$.

6.2 Small-core approximations

$$x_{\text{out}} = 1 - \epsilon^3 + O(\epsilon^6), \quad x_{\text{in}} = \epsilon^{3/2} + O(\epsilon^3) \quad (6.3)$$

The outer horizon remains close to r_s for small cores, while the inner horizon emerges deep inside. This separation explains why horizon-scale electromagnetic observables can remain close to general relativity even when the center is nonsingular.

6.3 Inner-horizon stability

The existence of a Cauchy horizon is a major unresolved issue. In many regular and charged black-hole geometries, counter-streaming perturbations produce mass inflation and large curvature near the inner horizon. Central regularity of a static metric is therefore not sufficient to establish dynamical regularity. The GCC completion succeeds only if the bounded-response field and backreaction prevent an unstable curvature blow-up during collapse and evaporation. This requires a time-dependent calculation and is a decisive falsification test.

7 Thermodynamics and a Conditional Evaporation Endpoint

7.1 Surface gravity and temperature

For the outer horizon, whose dimensionless radius is denoted by x_{out} , the surface gravity and Hawking temperature are

$$\kappa_{\text{out}} = \frac{c^2}{2r_s} f_x(x_{\text{out}}), \quad T_H = \frac{\hbar c}{4\pi k_B r_s} f_x(x_{\text{out}}) \quad (7.1)$$

$$f_x(x_{\text{out}}) = \frac{3x_{\text{out}} - 2}{x_{\text{out}}}, \quad \frac{T_H}{T_{\text{Sch}}} = \frac{3x_{\text{out}} - 2}{x_{\text{out}}} \quad (7.2)$$

The temperature approaches the Schwarzschild value as ϵ tends to zero and vanishes at the extremal point $x_{\text{out}} = 2/3$. This suppression is a property of the adopted static geometry; a full evaporation history additionally requires the response-field dynamics, greybody factors, backreaction, and a prescription for how r_c scales as the mass changes.

7.2 Conditional remnant scale

If r_c is a fixed microscopic length during evaporation, then ϵ grows as M decreases and the system reaches the extremal branch at

$$M_{\text{ext}} = \frac{c^2 r_c}{2G\epsilon_{\text{crit}}} \simeq 0.94494 \frac{c^2 r_c}{G} \quad (7.3)$$

For r_c equal to the Planck length this is of order the Planck mass. This is a conditional result, not a universal GCC prediction. If the core radius scales with mass or evolves dynamically, the endpoint can differ or disappear.

7.3 Entropy

When the Einstein-Hilbert term dominates at the outer horizon, the leading entropy is the usual area expression $S \approx k_B A_+/(4 l_P^2)$. The curvature coupling in Eq. (3.2) can generate Wald-entropy corrections. It would therefore be inconsistent to combine the full scalar-curvature action with an unmodified area law without computing the corresponding Noether charge.

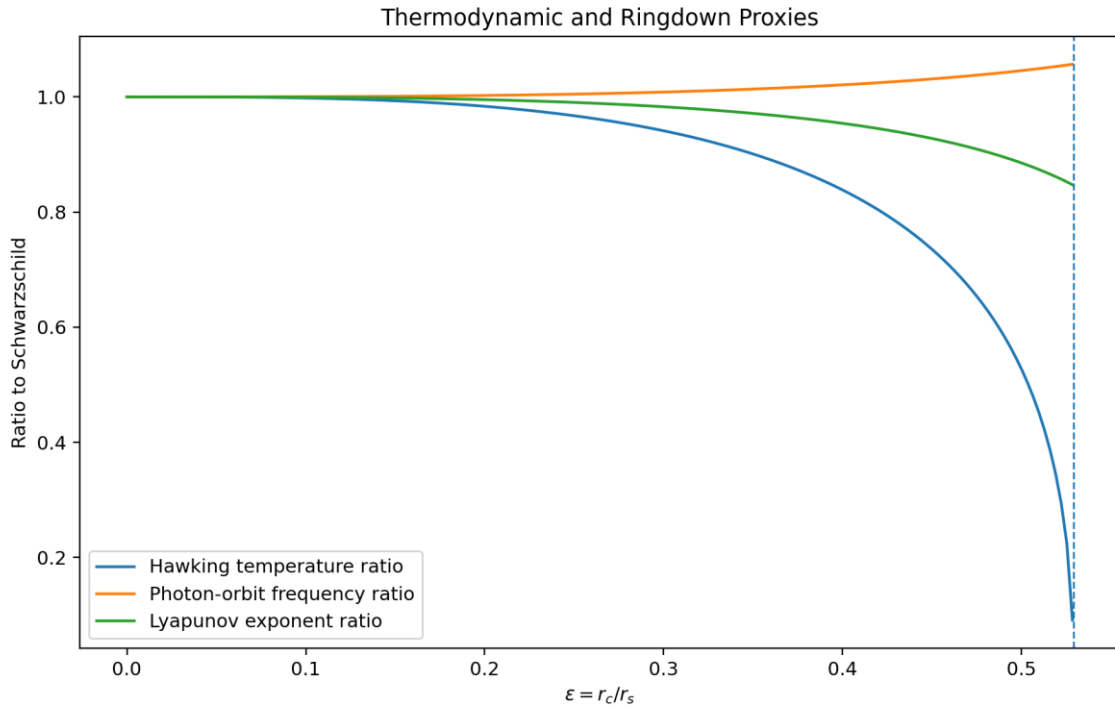


Figure 3. Hawking-temperature, photon-orbit-frequency, and Lyapunov-exponent ratios. The extremal branch is cold, while the photon-orbit frequency increases and the instability exponent decreases.

8 Photon Sphere, Shadow, and Circular Orbits

8.1 Photon sphere and shadow

The radius of an unstable circular null orbit satisfies

$$x_{\text{ph}} f_x(x_{\text{ph}}) = 2f(x_{\text{ph}}) \quad (8.1)$$

$$\frac{b_{\text{ph}}}{r_s} = \frac{x_{\text{ph}}}{\sqrt{f(x_{\text{ph}})}} \quad (8.2)$$

The Schwarzschild values are $x_{\text{ph}}=3/2$ and $b_{\text{ph}}/r_s=3\sqrt{3}/2$. The saturation core moves the photon sphere and shadow inward. The effect is small for ϵ much less than unity because the exterior correction begins at order ϵ cubed.

8.2 Timelike circular orbits and ISCO

$$\frac{L^2}{r_s^2} = \frac{x^3 f_x}{2f - x f_x}, \quad E^2 = \frac{2f^2}{2f - x f_x} \quad (8.3)$$

$$3f f_x + x f f_{xx} - 2x f_x^2 = 0 \quad (\text{ISCO}) \quad (8.4)$$

$$\frac{\Omega_{\text{ISCO}} r_s}{c} = \sqrt{\frac{f_x(x_{\text{ISCO}})}{2x_{\text{ISCO}}}} \quad (8.5)$$

The ISCO shifts inward and the characteristic orbital frequency rises. In a realistic accretion flow these geometric shifts are degenerate with spin, inclination, plasma emissivity, magnetic fields, and radiative transfer. The static values are therefore theory benchmarks, not direct observational fits.

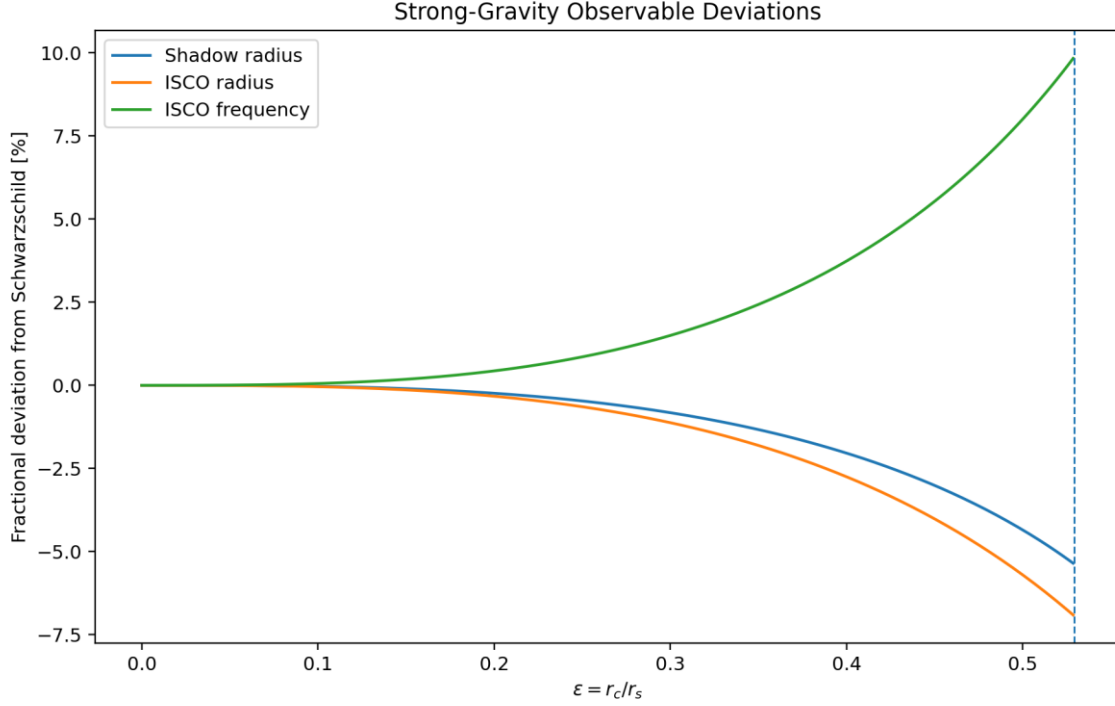


Figure 4. Fractional deviations of the shadow radius, ISCO radius, and ISCO orbital frequency from their Schwarzschild values. All quantities are fixed by the same ϵ .

8.3 Numerical benchmark

Table 4. Numerical benchmarks for the static saturation-core geometry.

epsilon	x_outer	b_ph/r_s	Shadow shift	x_ISCO	ISCO shift	Omega shift	T/T_Sch
0.0	1.0000	2.5981	+0.00%	3.0000	+0.00%	-0.00%	1.000
0.1	0.9990	2.5973	-0.03%	2.9988	-0.04%	+0.05%	0.998
0.2	0.9919	2.5919	-0.24%	2.9902	-0.33%	+0.43%	0.984
0.3	0.9714	2.5767	-0.82%	2.9663	-1.12%	+1.50%	0.941
0.4	0.9252	2.5451	-2.04%	2.9174	-2.75%	+3.74%	0.838
0.5	0.8090	2.4850	-4.35%	2.8289	-5.70%	+8.01%	0.528

9 Ringdown Proxy and Gravitational-Wave Tests

9.1 Eikonal diagnostic

For a single-field metric perturbation in the eikonal limit, the leading quasi-normal frequency is related to the unstable photon orbit:

$$\omega_{\ell p} \simeq \ell \Omega_{\text{ph}} - i \left(p + \frac{1}{2} \right) \lambda_{\text{ph}} \quad (9.1)$$

$$\frac{\Omega_{\text{ph}} r_s}{c} = \frac{\sqrt{f(x_{\text{ph}})}}{x_{\text{ph}}} \quad (9.2)$$

$$\frac{\lambda_{\text{ph}} r_s}{c} = \left\{ \frac{f(x_{\text{ph}})[2f(x_{\text{ph}}) - x_{\text{ph}}^2 f_{xx}(x_{\text{ph}})]}{2x_{\text{ph}}^2} \right\}^{1/2} \quad (9.3)$$

As epsilon increases, the real-frequency proxy rises while the Lyapunov exponent falls, implying slower eikonal damping. This trend is useful for screening parameter space, but it is not a full ringdown prediction. The scalar response field introduces additional perturbation degrees of freedom and can mix with metric modes. A complete calculation must derive the coupled axial and polar perturbation equations and test for ghosts, gradient instabilities, and growing modes.

9.2 Present observational status

Event Horizon Telescope images of M87* and Sagittarius A* show horizon-scale rings consistent with Kerr expectations within present astrophysical and calibration uncertainties. Gravitational-wave catalog tests likewise find no statistically significant departure from general relativity, and ringdown tests further restrict large deviations from Kerr. The present static model can therefore survive only in a regime where epsilon is modest or where spin-dependent degeneracies hide the spherical deviations. A quantitative constraint requires a rotating solution and source-specific forward modeling.

9.3 Conditional gravitational-wave echoes

A regular center does not automatically generate post-merger echoes. Echoes require a partially reflecting structure outside or effectively near the would-be horizon, or a horizonless ultracompact branch that forms a long-lived cavity. The two-horizon solution with epsilon below epsilon_crit still has an outer event horizon and therefore does not by itself return signals from the core.

If a completion-dependent saturation layer is located at $r_0 = r_+ (1 + \delta)$ and has nonzero frequency-dependent reflectivity $R(\omega)$, the leading round-trip delay is

$$\Delta t_{\text{echo}} \simeq \frac{2}{c} [r_{\text{tor}}(r_{\text{ph}}) - r_{\text{tor}}(r_0)] \simeq \frac{2r_s}{c} |\ln \delta| \quad (9.4)$$

where r_{tor} is the tortoise coordinate and the final expression is the near-Schwarzschild estimate for delta much smaller than unity. GCC presently predicts neither delta nor $R(\omega)$; therefore echo timing is a conditional scaling relation, while echo amplitudes and detailed waveforms remain open. A robust non-detection constrains reflective completions but does not falsify the regular core by itself.

10 Information Retention and the Information-Loss Problem

10.1 Microscopic unitarity and the absence of a singular sink

GCC postulates microscopic unitary updates. In that description, information is carried by the complete grid state and is not fundamentally destroyed by the coarse-grained emergence of a horizon. The classical singularity is problematic because it behaves as an endpoint where the effective continuum evolution ceases. Replacing it with a finite state region removes that terminal singular boundary and permits, in principle, continued microscopic evolution.

10.2 What regularity does not prove

A regular metric alone does not solve the information-loss problem. It does not calculate Hawking pair entanglement, the fine-grained radiation entropy, the Page time, the island prescription, or the transfer of interior correlations to outgoing modes. Nor does it establish that an extremal remnant has a finite number of internal states or avoids production and stability problems. The correct claim is narrower: a saturation core provides a kinematically available information-retention region and removes the singular endpoint, but the recovery channel remains dynamical and unproved.

10.3 Candidate recovery channels

Table 5. Candidate information-recovery channels.

Channel	Required calculation	Current status
Correlation-corrected Hawking radiation	Quantum fields on the dynamical GCC background and backreaction	Open
Late release near an extremal transition	Time-dependent evaporation solution and causal diagram	Conditional possibility
Long-lived remnant	State counting, stability, production bounds, entropy accounting	Conditional possibility
Horizon-to-horizon transfer through grid-microstates	Explicit microscopic update and coarse-graining map	Open
Echo or afterglow channel	Near-horizon reflectivity and coupled perturbation potential	Not implied by the static metric

11 Observational Interpretation and Parameter Strategy

11.1 Static nonrotating benchmarks

The parameter ϵ should first be treated as a dimensionless strong-field benchmark. For $\epsilon=0.4$, the shadow radius is smaller than Schwarzschild by about 2.0%, the ISCO radius by about 2.8%, the ISCO frequency is higher by about 3.7%, and the Hawking temperature is lower by about 16%. Near $\epsilon=0.5$, the corresponding changes become roughly 4-8% and the temperature is suppressed by nearly one half. These values demonstrate detectability scales without claiming a source-specific bound.

11.2 Need for rotation

Astrophysical black holes rotate. A useful GCC prediction must therefore provide a stationary axisymmetric completion with well-defined mass and current multipoles, horizon regularity, an unambiguous spin parameter, separability or a controlled numerical ray-tracing scheme, and coupled perturbation equations. Applying a Newman-Janis transformation mechanically is not sufficient unless the resulting stress tensor and field equations are physically consistent.

11.3 Multi-messenger no-retuning test

For a given object, one parameter set should predict the shadow morphology, orbital frequencies, iron-line and continuum observables, tidal response, merger waveform, ringdown spectrum, absorption, and any echoes. Agreement with one channel obtained by retuning the core separately in another channel would not constitute a test of the theory.

12 Falsifiability

Table 6. Strong-gravity falsification matrix.

Test	GCC saturation-core requirement	Failure condition
Field-theory consistency	Positive kinetic matrix; hyperbolic evolution; controlled curvature coupling	Ghost, gradient instability, or ill-posed Cauchy problem
Static regularity	Finite invariants and physically interpretable source	Hidden singularity or unacceptable stress tensor
Dynamical collapse	No mass-inflation singularity after backreaction	Generic divergent inner-horizon curvature
Exterior tests	One epsilon fits shadow, ISCO, lensing, and timing	Mutually inconsistent inferred core sizes
Ringdown	Stable coupled spectrum compatible with observations	Growing modes or excluded frequency/damping shifts
Weak-field recovery	Universal metric coupling and GR limit	Fifth force, preferred frame, or equivalence-principle violation
Information sector	Explicit unitary evolution and entropy accounting	Irreversible loss persists despite regularity
Cosmological consistency	Strong-field parameter does not spoil near- $n=3$ background	Large induced background exchange or CMB conflict

13 Discussion

13.1 Main achievement

The paper converts the open strong-field possibility of Paper I into a concrete one-parameter candidate. The geometry is regular, has a controlled Schwarzschild limit, supplies exact horizon and temperature formulas, and generates correlated deviations in several observables. It also makes clear that the successful cosmological correspondence is not used as evidence for the black-hole completion; the two regimes are separated.

13.2 Main limitation

The decisive missing result is a derivation of the mass profile from the nonlinear GCC dynamics. Without it, r_c is a phenomenological strong-field scale. The second major limitation is the inner Cauchy horizon, whose stability cannot be inferred from the static solution. The third is the absence of rotation. These are not cosmetic extensions: any one of them could invalidate the present completion.

13.3 Relation to established regular-black-hole models

Bardeen, Hayward, Dymnikova, and nonlinear-electrodynamic models already demonstrate that finite-density cores can replace the Schwarzschild singularity at the metric level. The mathematical novelty of Eq. (4.2) is therefore limited. The intended contribution is conceptual and programmatic: bounded response is tied to the discrete grid-clock substrate, all observables are organized under a no-retuning rule, cosmological constraints are carried forward honestly, and information recovery is not conflated with geometric regularity.

13.4 Effective grid activation and cosmological extensions

The local saturation variable used in this paper should not be identified automatically with a cosmological variation of the microscopic graph. Paper I distinguishes the fixed cardinality N_{grid} from a possible effective active count $N_{\text{eff}}(\tau, \mathbf{x})$. A future theory may relate strong-curvature saturation to a redistribution of active grid degrees of freedom, but the static solution derived here assumes

$$N_{\text{grid}} = \text{constant}, \quad n = 3 \quad (13.1)$$

and does not use N_{eff} to fit the homogeneous universe. Direction dependence, the H_0 tension, and cyclic evolution therefore remain cosmological research questions for Paper II-type extensions. Their inclusion would require a covariant field equation and a new likelihood; they are not consequences of the Hayward-type ansatz.

13.5 Recommended development sequence

- Derive a static regular solution directly from a ghost-free bounded-response action and identify r_c in terms of action parameters.
- Perform a full linear stability analysis, especially at the inner horizon.
- Construct a rotating solution and calculate shadows, multipoles, ISCOs, and horizon absorption.
- Derive coupled quasi-normal modes and compare with gravitational-wave ringdown constraints.
- Build a dynamical collapse-and-evaporation solution and compute fine-grained information diagnostics.
- Fit all strong-gravity observables with a shared parameter set and without changing the cosmological closure.

14 Conclusion

Grid-Clock Cosmology III proposes a bounded-response interpretation of regular black holes. The minimum saturation-core metric is Schwarzschild-like outside, de Sitter-like inside, and regular at the center. Its causal structure is governed by $\epsilon = r_c/r_s$, with two horizons below $\epsilon_{\text{crit}} = (4/27)^{1/3}$, an extremal cold horizon at the critical value, and a horizonless branch above it. The same ϵ predicts inward shifts of the photon sphere, shadow, and ISCO, an increased orbital frequency, a suppressed Hawking temperature, and characteristic eikonal ringdown trends.

The construction is falsifiable but incomplete. It does not yet derive the regular mass profile from the microscopic grid update, resolve inner-horizon stability, provide a rotating metric, or calculate unitary evaporation. Its scientifically defensible claim is therefore not that the black-hole information problem has been solved, but that GCC admits a quantitatively defined strong-field completion candidate in which the singular endpoint is replaced by finite continued dynamics.

Together, the three GCC papers establish a coherent hierarchy: a discrete substrate and correspondence structure, a minimum cosmological model, a covariant CMB and growth test that returns the homogeneous sector to Λ CDM limit, and a separate strong-field completion with explicit observables and failure conditions. Possible time-, load-, or direction-dependent running of the cosmological response index is reserved for future work and is not used to obtain the black-hole result. The final verdict on the black-hole sector must come from nonlinear derivation, stability, rotation, and multi-messenger observation.

Appendix A. Useful Dimensionless Equations

Table A1. Useful dimensionless equations.

Quantity	Equation
Metric function	$f(x) = 1 - \frac{x^2}{x^3 + \epsilon^3}$
Horizon equation	$x^3 - x^2 + \epsilon^3 = 0$
Critical core	$\epsilon_{\text{crit}} = \left(\frac{4}{27}\right)^{1/3} = 0.529133684$
Photon sphere	$x_{\text{ph}} f_x(x_{\text{ph}}) = 2f(x_{\text{ph}})$
Shadow radius	$\frac{b_{\text{ph}}}{r_s} = \frac{x_{\text{ph}}}{\sqrt{f(x_{\text{ph}})}}$
Circular angular momentum	$\frac{L^2}{r_s^2} = \frac{x^3 f_x}{2f - x f_x}$
ISCO condition	$3f f_x + x f f_{xx} - 2x f_x^2 = 0$
Orbital frequency	$\frac{\Omega_{\text{ISCO}} r_s}{c} = \sqrt{\frac{f_x}{2x}}$
Temperature ratio	$\frac{T_H}{T_{\text{Sch}}} = \frac{3x_{\text{out}} - 2}{x_{\text{out}}}$
Eikonal QNM proxy	$\omega_{\ell p} \simeq \ell \Omega_{\text{ph}} - i \left(p + \frac{1}{2}\right) \lambda_{\text{ph}}$

Appendix B. Central Stress-Energy and Curvature

For the effective mass profile $m(r) = \frac{Mr^3}{r^3+r_c^3}$, the density is finite and maximal at the center, with $\rho(0) = \frac{3M}{4\pi r_c^3}$.

The central equation of state is $p_r = -\rho$. The exact Ricci scalar is

$$R(r) = \frac{12Mr_c^3(2r_c^3-r^3)}{(r^3+r_c^3)^3} \quad (\text{B.1})$$

The finite center follows immediately. The sign change of R away from the core reflects the anisotropic effective source rather than a second singularity.

Appendix C. Prediction Hierarchy for the Black-Hole Sector

Table C1. Prediction hierarchy for the black-hole sector.

Tier	Prediction class	Examples
1	Mandatory correspondence	Schwarzschild/Kerr recovery at weak curvature; universal free fall
2	Static ansatz predictions	Horizons, shadow, ISCO, surface gravity as functions of epsilon
3	Action-dependent predictions	Scalar hair, Wald entropy, complete QNMs, tidal Love numbers
4	Microscopic completion predictions	Core state counting, unitary evaporation, Page curve, bounce or remnant dynamics

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